Tomás Berriel Martins

berriel.tomas@gmail.com • tberriel.github.io • in tomas-berriel

3D Perception Engineer

Professional Summary

PhD researcher specializing in 3D scene understanding, neural rendering, and open-vocabulary semantic mapping, with 4 peer-reviewed publications at RA-L+ICRA, IROS, BMVC, and ICCVW+3DV, and one currently under review. Recent Student Researcher at Google's AR Semantic Perception team (Munich), working on IMU-integrated 3D multi-view reconstruction. Seeking a Research Engineer or 3D Vision role at the intersection of neural rendering, semantic perception, and real-world deployment. PhD defense expected December 2026.

Technical Skills

Languages: Python, C++

Frameworks & Tools: PyTorch, Lightning, NumPy, Hydra, Git, SLURM

Research areas: 3D Gaussian Splatting, Neural Radiance Fields (NeRF), Visual SLAM, Open-Vocabulary Semantic Mapping, Multiview Geometry, Vision–Language Models

Publications

- **[a1] Cross-Attentive Multiview Fusion of Vision-Language Embeddings.** Tomás Berriel Martins, Martin R. Oswald, Javier Civera. Under Review, 2026.
- ***[c4]* SLAM&Render: A Benchmark for the Intersection Between Neural Rendering, Gaussian Splatting and SLAM.** Samuel Cerezo, Gaetano Meli, Tomás Berriel Martins, Kirill Safronov, Javier Civera. IEEE IROS, 2025.
- ***[j1-c3]* Open-Vocabulary Online Semantic Mapping for SLAM.** Tomás Berriel Martins, Martin R. Oswald, Javier Civera. IEEE Robotics and Automation Letters, 2025; IEEE ICRA, 2026.
- ***[c2]* Feature Splatting for Better Novel View Synthesis with Low Overlap.** Tomás Berriel Martins, Javier Civera. BMVC, 2024.
- ***[w1-c1]* Ray-Patch: An Efficient Querying for Light Field Transformers.** Tomás Berriel Martins, Javier Civera. ICCV Workshops 2023; 3DV 2024.

Work & Research Experience

Google **Munich, Germany**
Student Researcher Intern *Oct 2025–Apr 2026*
AR Semantic Perception team. Researched the integration of IMU data to improve 3D multi-view scene reconstruction, combining inertial sensing with vision-based neural representations. Supervised by Juan José Gómez.

University of Zaragoza – RoPeRT Group **Zaragoza, Spain**
Predoctoral Researcher *2022–2026*

Full-time PhD research on 3D scene understanding: designed and built an online open-vocabulary semantic mapping system for SLAM (RA-L 2025, ICRA 2026); developed a 3D Gaussian Splatting method for novel view synthesis under low overlap (BMVC 2024); and researched cross-attentive vision-language feature fusion for multiview 3D reconstruction (under review). Advisor: Prof. Javier Civera.

University of Amsterdam – Computer Vision Group **Amsterdam, Netherlands**
Visiting Researcher *2024 (4 months)*
Built an online pipeline for open-vocabulary semantic 3D reconstruction integrating vision-language embeddings with neural scene representations, resulting in the RA-L 2025 publication. Supervised by Asst. Prof. Martin R. Oswald.

University of Zaragoza – Robotics, Perception & Real Time Group **Zaragoza, Spain**
Research Engineer *2020–2021*
Developed Bayesian Neural Networks for uncertainty-aware 360° indoor layout estimation from single panoramic images. Advisor: Prof. Javier Civera.

ITAinnova **Zaragoza, Spain**
R&D Engineer Intern *2019–2020*
Built perception and navigation software in C++/Python with ROS for autonomous indoor and outdoor platforms within a multidisciplinary robotics team. Supervisor: Javier Huarte.

Education

University of Zaragoza **Spain**
PhD in Systems Engineering and Computer Science *2021–2026 (expected Dec)*
Thesis: 3D scene understanding via neural representations and open-vocabulary semantic mapping. Advisor: Prof. Javier Civera.

University of Zaragoza **Spain**
Master in Robotics, Graphics, and Computer Vision *2020–2022*
Thesis: *Learning disentangled representations of scenes from images*. Advisor: Prof. Javier Civera.

University of Zaragoza **Spain**
Bachelor's Degree in Electronic and Automatic Engineering *2015–2019*
Thesis: *Automated human actions recognition in 3D video sequences*. Advisor: Prof. Carlos Orrite.

Teaching

2021–2025: Teaching Assistant (180 h total) for Computer Vision, Machine Learning, SLAM, and Deep Learning courses at Master's and Bachelor's level, University of Zaragoza.

Languages

English: Fluent
Italian: Fluent
Spanish: Native